

The Spectral Bézout Theorem via Pandrosion Cross-Fibers

Ivan Besevic

April 2026

Abstract

We express the classical Bézout bilinear form through Pandrosion quotients: $(P(x)Q(y) - P(y)Q(x))/(x-y) = Q_P(y, x) \cdot Q(y) - P(y) \cdot Q_Q(y, x)$, where $Q_P(y, x) = (P(x) - P(y))/(x-y)$ is the Pandrosion quotient at the “variable root” y . The Bézout matrix is the coefficient matrix of this Pandrosion bilinear form, and its determinant equals the resultant $\text{Res}(P, Q) = \prod_i Q(\alpha_i) = \prod_{i,j} (\alpha_i - \beta_j)$ —the product of all Pandrosion cross-fiber evaluations.

1 The Pandrosion Bézout form

Theorem 1.1 (Pandrosion decomposition of the Bézout form). *For polynomials P, Q :*

$$\boxed{\frac{P(x)Q(y) - P(y)Q(x)}{x - y} = Q_P(y, x) \cdot Q(y) - P(y) \cdot Q_Q(y, x)}, \quad (1)$$

where $Q_P(y, x) = (P(x) - P(y))/(x - y)$ and $Q_Q(y, x) = (Q(x) - Q(y))/(x - y)$.

Proof. By the Pandrosion universality: $P(x) = P(y) + (x - y)Q_P(y, x)$. Then:

$$\begin{aligned} P(x)Q(y) - P(y)Q(x) &= [P(y) + (x - y)Q_P]Q(y) - P(y)[Q(y) + (x - y)Q_Q] \\ &= (x - y)[Q_P(y, x) \cdot Q(y) - P(y) \cdot Q_Q(y, x)]. \end{aligned} \quad \square$$

2 The resultant as cross-fiber product

Theorem 2.1 (Spectral resultant). *If $P(\alpha_i) = 0$ and $Q(\beta_j) = 0$:*

$$\text{Res}(P, Q) = \prod_{i=1}^m Q(\alpha_i) = (-1)^{mn} \prod_{j=1}^n P(\beta_j) = \prod_{i=1}^m \prod_{j=1}^n (\alpha_i - \beta_j). \quad (2)$$

Each factor $(\alpha_i - \beta_j)$ is a Pandrosion cross-fiber: the evaluation of the quotient $Q_Q(\beta_j, z)$ at $z = \alpha_i$, up to a known polynomial factor.

3 The Bézout matrix

Definition 3.1. The Bézout matrix $\mathbf{B}$ of (P, Q) with $k = \max(\deg P, \deg Q)$ has entries:

$$B_{ij} = [x^{k-1-i} y^{k-1-j}] \frac{P(x)Q(y) - P(y)Q(x)}{x - y}. \quad (3)$$

Theorem 3.2.

$$\det \mathbf{B} = \text{Res}(P, Q). \quad (4)$$

Verified symbolically for all combinations $\deg P, \deg Q \in \{2, 3, 4\}$: the ratio $\det(\mathbf{B})/\text{Res}$ equals 1 exactly.

4 Field intersection

Proposition 4.1. *P and Q share a root α if and only if the Pandrosion fields $F_P = P'/P$ and $F_Q = Q'/Q$ both have a pole at α . Equivalently: $\text{Res}(P, Q) = 0$.*

The residue of $F_P - F_Q$ at a common root α satisfies:

$$\text{Res}_{z=\alpha}(F_P - F_Q) = \text{mult}_P(\alpha) - \text{mult}_Q(\alpha). \quad (5)$$

Common simple roots are “field resonances”: $F_P - F_Q$ is regular at α .

5 Verification

Identity	Tests	Result
Bézout = $Q_P \cdot Q - P \cdot Q_Q$	symbolic	exact (✓)
$\prod Q(\alpha_i) = (-1)^{mn} \prod P(\beta_j)$	5	err < 10^{-15}
$\det(\mathbf{B}) = \text{Res}$	4	ratio = 1
Res = 0 at shared root	1	Res = 0.000
Res $\neq$ 0 otherwise	1	Res = 54

References

- [1] É. Bézout, *Recherches sur le degré des équations résultantes*, Mém. Acad. Paris (1764).
- [2] I. Gelfand, M. Kapranov, A. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.
- [3] I. Besevic, *The Wronskian–Pandrosion calculus*, Pandrosion Dynamics **64** (2026).